A differentiable finite-element solver for superconducting-qubit design: converged frequencies and gradient re-optimization of the DeviceLayout.jl transmon

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Abstract

The SingleTransmon example of DeviceLayout.jl publishes the qubit and resonator frequencies of an initial and an optimized transmon design, computed with Palace at DeviceLayout's second-order discretization. We cross-check them with Palace and with geode-fem, an independently implemented finite-element solver with design derivatives. Palace at DeviceLayout's discretization reproduces both published pairs to within 0.7%, and on shared meshes the two solvers agree on the qubit mode to 10^{-4}%. Palace's resolution ladders, whose order-1 levels at h and $h/2$ geode-fem reproduces, extrapolate to qubit frequencies 7–9% above the published 4.14 and 3.005 GHz and to resonator frequencies 2.7% above. The finest computed levels already lie 4.1% and 4.7% above the published qubit values, and adaptive refinement and a second refinement family of the initial design fall inside the extrapolation band. A certified bracket on the island capacitance with the sapphire tensor, $[79.56, 79.76]$ fF, read through a two-node circuit, gives an initial-qubit estimate of 4.45 GHz and, under that circuit model, a cap of 4.477 GHz. Newton steps whose Jacobian combines geode-fem's capacitor-length shape derivative with a transmission-line model meet DeviceLayout's stopping criterion in 3 Palace runs, where its derivative-free optimizer used 9; a scaling-law Jacobian also needs 3, so the saving comes from Newton's method, not from the derivative. Aimed at converged frequencies, the iteration finds that the 3.0/4.0 GHz design needs a capacitor about 19% longer (16–38% within the extrapolation band) and a resonator 2.4% longer. Converged frequencies are extrapolations with stated bands, not bounds; only the capacitance brackets are certified.

1 Introduction

Superconducting-qubit layouts are designed by iterating field solves: a designer parameterizes a geometry, simulates it, compares the extracted frequencies or Hamiltonian parameters with targets, and adjusts. The state-of-practice automated workflow, QDesignOptimizer (Eriksson et al., 2025), makes the loop explicit and steers it with analytic models because its field solver returns observables but no derivatives of them. Open tools now cover the whole loop. DeviceLayout.jl (AWS Center for Quantum Computing, 2025a) builds the layout and its mesh, and Palace (Palace developers, AWS Center for Quantum Computing, 2023), an MFEM-based solver (Anderson et al., 2021; Andrej et al., 2024), computes the eigenmodes. DeviceLayout's SingleTransmon example (AWS Center for Quantum Computing, 2025b; Peairs and Carson, 2025) runs the loop end to end. It publishes an initial transmon-plus-resonator design at 4.14 and 5.591 GHz and a design optimized to targets of 3.0 and 4.0 GHz, at 3.005 and 3.982 GHz, found with a derivative-free optimizer in nine Palace runs.

Those frequencies come from one discretization, and the example itself warns that they are sensitive to it. This paper measures how sensitive. Our main result is that the published frequencies

are discretization-limited: Palace's resolution ladders extrapolate to qubit frequencies 7–9% above them, geode-fem reproduces the two ladder levels it can reach to 10^{-4}%, adaptive refinement and a second refinement family fall inside the extrapolation band, and a certified bound on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model. A Newton re-optimization at converged resolution then estimates that the optimized design needs a capacitor about 19% longer (an extrapolated estimate, 16–38%).

The second solver is GEODE-FEM ("geode-fem"), a Rust finite-element code on the Burn tensor framework (Tracel AI, 2024). It solves the same $H(\mathrm{curl})$ eigenproblem as Palace with an independent implementation, and it computes design derivatives: shape and material sensitivities of its electrostatic and driven operators, and the Hellmann–Feynman shape derivative of an eigenfrequency. On the committed DeviceLayout mesh, given as the same input file to both codes, geode-fem reproduces Palace's six eigenmodes above 4.5 GHz to $\leq 0.033\%$ and the qubit mode to 5×10^{-5} % (Section 4). This agreement establishes that the two codes assemble and solve the same discrete problem. It also gives an independent same-mesh cross-check of the Palace/DeviceLayout.jl transmon stack, complementing the earlier checks of Palace by SQDMetal (Sommers et al., 2025) and Ye et al. (2025).

Our contributions are:

- **A converged cross-check of a published design example.** Both published geometries are meshed by DeviceLayout itself and solved on a resolution ladder with first- and second-order elements. Palace at DeviceLayout's discretization reproduces the published pairs to under 0.7%. The converged estimates lie 7.3% and 8.7% above the published qubit frequencies and 2.7% above the resonators, and every published value lies below its extrapolation band. Palace adaptive refinement and a second refinement family of the initial geometry are consistent with the estimate (Section 5).

- **Certified capacitance brackets with the sapphire tensor.** A conforming upper bound and an equilibrated-flux lower bound, a classical pairing applied here to an island capacitance with a floating feedline, bracket the converged island capacitance of the unmodified device in $[79.56, 79.76]$ fF. Read through the junction and resonator circuit they give an initial-qubit estimate of 4.45 GHz, inside the eigenmode band, and a cap of 4.477 GHz that holds under the circuit model. The certified difference between the sapphire tensor and its scalar average is 2.5–3.1% (Section 6).

- **A Newton re-optimization with controls.** Newton steps in DeviceLayout's own variables, with a Jacobian from geode-fem's eigenfrequency shape derivative and a transmission-line model, meet DeviceLayout's criterion in 3 Palace runs against the published 9. A control with a scaling-law Jacobian and no geode-fem solve also takes 3, and Broyden's method 5, so the run count measures Newton's method in these variables, not the shape derivative. At converged resolution the iteration moves the optimized capacitor from 1217.678 to about 1446 μm, an extrapolated estimate whose band spans 16–38% over the published length (Section 7).

- **A validated sensitivity layer.** Material and geometry derivatives of the electrostatic and driven $H(\mathrm{curl})$ operators, each checked against finite differences of the full pipeline with mutation tests (Section 8).

- **Same-input-mesh agreement with Palace** on six eigenmodes and the qubit mode, with the 17.49 GHz mode that both codes share identified as an artifact of the distributed port model. It is an independent same-mesh cross-check of the Palace/DeviceLayout.jl transmon

stack, complementing the earlier checks of Palace by SQDMetal (Sommers et al., 2025) and Ye et al. (2025).

Section 2 places the work. Section 3 describes the solvers, the device and the models; Section 4 the same-mesh agreement; Sections 5–7 the cross-check, the certified capacitance and the re-optimization; Section 8 the sensitivity layer; Section 9 how geode-fem was designed and built; and Section 11 what is and is not established. Appendix A reports an electrostatic design loop with a certified acceptance test, and Appendix B the performance context.

2 Related Work

Design sensitivities in finite-element electromagnetics. Adjoint and direct sensitivity analysis of frequency-domain electromagnetic solvers is an established field. Nikolova et al. (2004) review adjoint-variable methods for high-frequency structures across FEM, MoM, TLM and FDTD. On tetrahedral vector elements, Webb and coworkers derived design sensitivities with high-order vector elements and of the frequency response, and used them to optimize microwave devices (Nair and Webb, 2003; Webb, 2001, 2002), and continuum shape sensitivity for high-frequency devices has been formulated as well (Choi et al., 2011). Shape adjoints for Maxwell cavity eigenvalues were developed for accelerator cavities, for shape optimization (Akçelik et al., 2005) and for inverse shape determination of deformed cavities (Akçelik et al., 2008), and a recent study optimizes Maxwell eigenvalues by shape variations with hand-derived adjoint gradients (Herter et al., 2026). Commercial solvers expose derivatives: HFSS computes analytic S-parameter derivatives with respect to geometric and material variables by reusing the forward LU factors (Ansys, Inc., 2025; Vardapetyan et al., 2008), and COMSOL offers forward and adjoint sensitivities and electromagnetic shape optimization (COMSOL AB, 2023; Jensen, 2022). Automated shape differentiation for general FEM exists in UFL/Firedrake (Ham et al., 2019) and NGSolve (Gangl et al., 2021). Palace exposes no derivatives. For capacitance, adjoint sensitivities are standard in interconnect extraction (El-Moselhy et al., 2008) and FEM (Ren et al., 2012), and the result that the shape derivative of a stored-energy capacitance reduces to its explicit term is the classical self-adjoint case (Park, 2019). What geode-fem adds is an open implementation of these derivatives next to its solver, applied here to a qubit design loop; the methods are not new.

Superconducting-qubit design optimization. QDesignOptimizer (Eriksson et al., 2025) wraps non-differentiable simulation with physics-guided updates. The SQcircuit line (Rajabzadeh et al., 2023, 2024) and SuperGrad (Wang et al., 2025) differentiate at the circuit-Hamiltonian level, so the map from geometry to circuit parameters still runs through field simulation. SQuADDS (Shanto et al., 2024) replaces simulation by lookup in a validated design database, and Eun et al. (2023) shape-optimize transmon pads with FEM and a derivative-free global optimizer. Qiskit Metal's quasi-lumped methods (Minev et al., 2021b) and energy-participation and black-box quantization (Koch et al., 2007; Minev et al., 2021a; Nigg et al., 2012) supply the parameter extraction. To our knowledge, within qubit EM design no prior work drives a layout with gradients from the field solve. DeviceLayout's example optimizes with PRIMA (Zhang, 2023), a reference implementation of Powell's model-based derivative-free methods (Powell, 1994, 2006), which build an interpolation model of the objective from function values.

Certified two-sided capacitance bounds. Bracketing a capacitance between complementary energy principles is classical, from the analytic variational bounds of Polya and Szegö (1951) to finite-element dual formulations (Hammond and Penman, 1976; Penman and Fraser, 1983), and in

three dimensions Specogna (2011) gives complementary formulations that bound capacitance from both sides. Our lower bound uses the hypercircle idea of Prager and Synge (1947) with Raviart–Thomas fluxes (Raviart and Thomas, 1977) equilibrated patch by patch (Braess and Schöberl, 2007; Braess et al., 2009; Ern and Vohralík, 2015), the route of guaranteed goal-oriented bounds such as Parés et al. (2021). The method is not new. What we add is its application to an island capacitance with a floating conductor on a production qubit layout, used as a design-acceptance test; we found no certified capacitance bound in the qubit-design literature.

Cross-checks of Palace. SQDMetal (Sommers et al., 2025) cross-checks a Palace workflow against COMSOL and Ansys with mesh-convergence studies, and Ye et al. (2025) compare a Palace workflow with cryogenic measurement at the $\sim$0.3% level. Both use models built independently for each tool. Our comparison instead holds the input mesh and the junction model fixed and changes the solver implementation, and it adds a converged reference for a published design. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997) it is code verification and code-to-code comparison, not validation against measurement. Code-to-code benchmarking on precisely specified problems is a long tradition in computational electromagnetics (Hoffmann et al., 2009). Palace and DeviceLayout.jl have no methods publication (AWS Center for Quantum Computing, 2025a; AWS Quantum Technologies Blog, 2023; Palace developers, AWS Center for Quantum Computing, 2023; Peairs, 2025; Peairs and Carson, 2025).

Differentiable FEM on general tensor stacks. Differentiable electromagnetics is most developed in photonics, on structured grids and integral methods (Hammond et al., 2022; Hughes et al., 2019; Kim et al., 2024; Mahlau et al., 2026; Molesky et al., 2018; Ponomareva et al., 2025; Schubert et al., 2022). Differentiable FEM on autodiff tensor stacks exists outside EM (Liang et al., 2023; Xue et al., 2023) and for general Galerkin assembly (Chi and Wen, 2026; Wen et al., 2026). FEM kernel generation is a long tradition (Alnæs et al., 2014; Homolya et al., 2018; Kirby and Logg, 2006; Rathgeber et al., 2017), and its GPU form is libCEED (Brown et al., 2021), on which Palace runs. geode-fem phrases assembly as batched tensor algebra on a general-purpose ML stack (Chen et al., 2018; Ragan-Kelley et al., 2013). The design derivatives reported here do not come from that stack's autodiff: they use hand-written forward-mode twins of the element kernels and a discrete adjoint (Section 3.1).

3 Solvers, Device and Models

3.1 geode-fem

geode-fem assembles first-order Nédélec (Nédélec, 1980) and scalar P1/P2 element operators as batched $[n_{\text{elem}}, k, k]$ tensor operations on Burn, whose cubecl compiler targets CPU, CUDA and WebGPU from one source. Its eigensolver is shift-invert Lanczos (Lehoucq et al., 1998) on a real symmetric pencil, factorized with faer (Kazdadi, 2026); large electrostatic solves use conjugate gradients with an algebraic-multigrid preconditioner in native f64. Burn's autodiff tape reaches the assembled operators but not the sparse factorization, and the derivatives in this paper do not use the tape. A discrete-adjoint layer supplies the derivative through the solve with one transpose solve on the same factorization (Section 8). The eigenfrequency derivative used in Section 7 needs no adjoint solve, because the pencil is symmetric: for a simple eigenpair (λ, x) with $x^{\mathsf{T}} M x = 1$,

$$\frac{d\lambda}{d\theta} \;=\; x^{\mathsf{T}}\Big(\frac{\partial K}{\partial \theta} - \lambda\,\frac{\partial M}{\partial \theta}\Big)x, \tag{1}$$

the Hellmann–Feynman (Nelson) formula (Nelson, 1976). Here K and M are the full matrices of the pencil, including the junction terms of Eq. (2); the junction does not move under the capacitor-length motion of Section 7, so those terms do not contribute. geode-fem evaluates the element derivatives $\partial K/\partial\theta$ and $\partial M/\partial\theta$ along a node velocity $\partial X/\partial\theta$ with a hand-written forward-mode dual-number twin of its tensor-permittivity Nédélec kernel, whose real part is tested against the assembled kernel to 10^{-10} relative.

3.2 Device, mesh and materials

The device is DeviceLayout.jl v1.15.0's SingleTransmon example (AWS Center for Quantum Computing, 2025a): a transmon with a meander readout resonator and a feedline on sapphire, in a vacuum box (Figure 1). Its two design parameters are the transmon's `capacitor_length` c and the resonator's `total_length` L; the published initial design has $c = 620\,\mu$m and $L = 5000\,\mu$m. The sapphire permittivity is $\mathrm{diag}(9.3, 9.3, 11.5)$ with the optic axis normal to the chip. The configuration rotates the material axes about the chip normal by about 36.87°; for a uniaxial tensor whose optic axis is that normal the rotation is the identity, so the lab tensor is diagonal. Where a scalar model is used it is the trace average $\bar{\varepsilon} = 10.03$.

Two mesh families appear. The committed fixture is DeviceLayout's mesh of the initial design (22,684 nodes, 133,314 tetrahedra), pinned by SHA-256; Section 4 and the capacitance brackets of Section 6 use it and its refinements, which leave its polygonal surfaces unchanged. For the cross-check and the re-optimization, DeviceLayout itself meshes each geometry with the two parameters exposed, at first order for the ladders and at its own second order for the runs that reproduce its discretization. DeviceLayout's second-order mesh has the same tetrahedra as its first-order mesh with curved faces; on the same element order, curving changes the qubit by at most 0.0001% and the resonator by about 0.05%. On the initial geometry the two families give the same Palace qubit at h (3.4527 against 3.4520 GHz, 0.02%). PEC is applied on the metal sheets and the box; the readout ports are left open in every eigenproblem except the DeviceLayout-configuration runs of Section 5.

3.3 Junction and circuit models

The Josephson junction is a linear lumped element, $L_J = 14.860\,$nH in parallel with $C_J = 5.5\,$fF, on the four-triangle sheet Γ of Figure 1b, exactly as in DeviceLayout's Palace configuration. With port length ℓ and width w the uniform lumped-port construction adds

$$K_{\mathrm{port}} = \frac{\ell}{w\,L_J}\,S_\Gamma \qquad \text{and} \qquad M_{\mathrm{port}} = \frac{C_J\,\ell}{w}\,S_\Gamma \tag{2}$$

to the pencil $(K + K_{\mathrm{port}})x = \omega^2(M + M_{\mathrm{port}})x$, where S_Γ is the isotropic surface matrix on Γ. Palace's `LumpedPort` adds the same terms with the same scalings, so the default models of the two codes coincide. The isotropic sheet also stores energy in a zero-voltage field pattern confined to the patch, which produces a mode at 17.49 GHz in both codes (Section 4). geode-fem's rank-1 port-voltage shunt replaces S_Γ by $b\,b^{\mathsf{T}}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\,dS$ the port-voltage functional. It removes that mode and keeps the others to four decimals on the fixture, and it is the junction model of every geode-fem eigensolve in Sections 5 and 7.

The electrostatic model splits the metal into three conductors: ground with the resonator, the floating feedline, and the island. Its island capacitance with the feedline floating and neutral, $C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$, excludes C_J. A two-node circuit connects it to the eigenproblem: a bare qubit $1/(2\pi\sqrt{(L_J + L_s)(C_\Sigma + C_J)})$, with L_s the series inductance of the junction's current path, coupled with coefficient k to the resonator. On the fixture L_s and k were measured on three and two discretizations; extrapolated, they give bands of 0.113–0.125 nH and 0.0227–0.0310, which are estimates, not bounds.

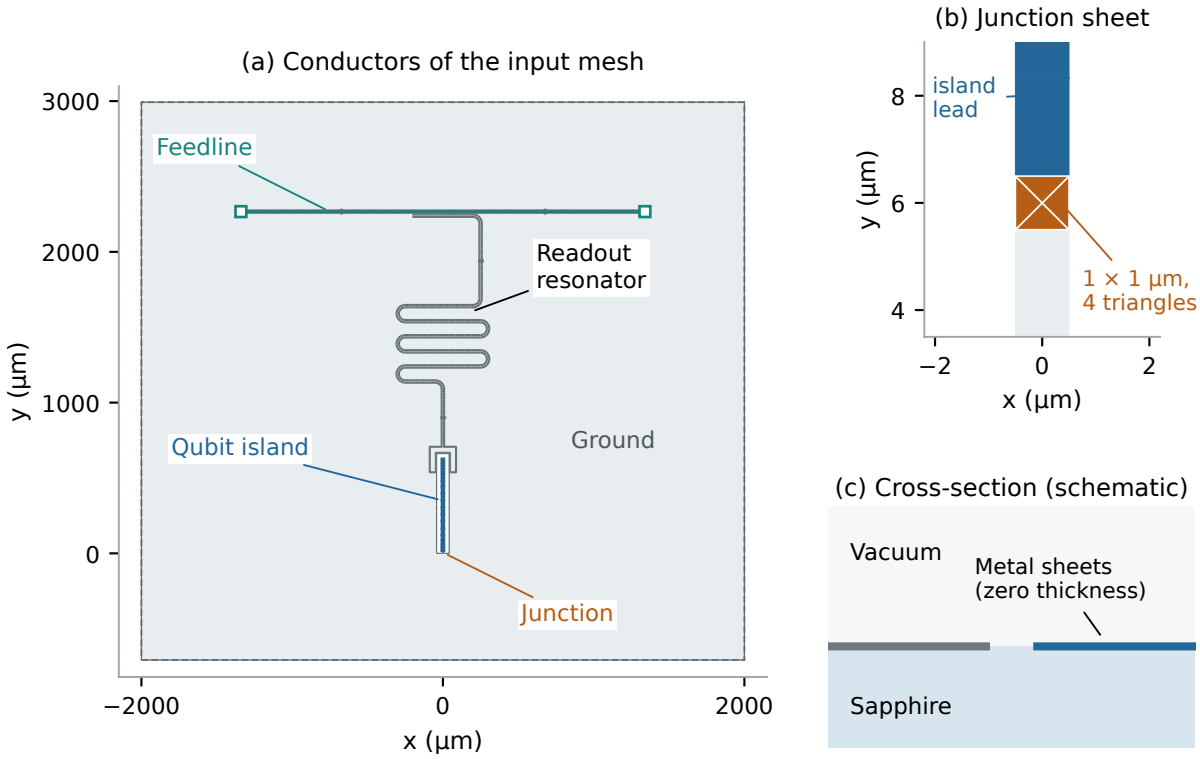


Figure 1: Benchmark geometry, drawn from the committed SHA-256-pinned MSH 4.1 input mesh of the DeviceLayout.jl v1.15.0 `SingleTransmon` device. (a) Top view of the three node-disjoint metal conductors of Section 6: the feedline (teal), the qubit island (blue), and the ground plane together with the meander readout resonator (gray), whose etched gaps are drawn as dark outlines. Squares mark the two readout ports; the dashed line is the exterior-boundary outline. (b) The $1 \times 1\,\mu$m junction `lumped_element` sheet with its four-triangle triangulation. (c) Material cross-section, schematic and not to scale; the `substrate` and `vacuum` volume groups fill the enclosing box. Both solvers read this file (22,684 nodes; 133,314 tetrahedra); Palace then refines it internally around the metal sheets (Section 4).

4 Same-Input-Mesh Agreement With Palace

Both codes read the committed fixture with the same materials and junction model at first order. Table 1 gives the result. Palace returns the qubit mode only when its search target lies below it, because its eigensolver searches above the target (read from Palace's source and configuration schema at the commit of Section 10); the six modes above 4.5 GHz come from a run with `Target` = 4.5 GHz and the qubit from one with `Target` = 2 GHz. All six agree to $\leq 0.033\%$ (worst case 0.032%), and the qubit mode to 5×10^{-5} %, with port participation 0.9942 in geode-fem and port EPR 0.99424 in Palace. Doubling L_J moves the qubit by a ratio of 0.70810 in Palace and 0.7081 in geode-fem, close to $1/\sqrt{2}$, and a Palace rerun from the committed configuration reproduces its `eig.csv` bit for bit.

The 17.49 GHz mode has an explanation that is an inference, which a synthetic test supports. In geode-fem all its stiffness energy is in the port term ($p = 1.000$), while Palace, which builds its port EPR from the port voltage, gives it 2.49×10^{-8}; its energy sits in a surface pattern with zero net voltage. The junction patch has one free node, whose hat-function gradient carries no

Table 1: Eigenmode agreement from the identical input mesh (22,684 nodes, 133,314 tetrahedra), first order in both solvers, junction as the lumped $L_J \parallel C_J$ shunt. geode-fem solves 156,863 Nédélec unknowns (133,108 interior); Palace refines the mesh internally around the metal sheets and solves 183,891. geode-fem frequencies are the committed three decimals (six for the qubit), and Δ is the committed relative difference. These are same-mesh values, not converged frequencies. Source: `benchmarks/transmon_eigen/results.toml`.

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
qubit	transmon qubit	3.452751	3.452749306805	0.00005
resonator	readout resonator	5.153	5.151335830348	0.032
mode 2	box/cavity	15.465	15.46052107794	0.029
port sheet	port-sheet artifact	17.490	17.49010903536	0.001
mode 4	box/cavity	18.693	18.69165792915	0.007
mode 5	box/cavity	20.703	20.69755679425	0.026
mode 6	box/cavity	26.088	26.08089940472	0.027

port voltage but still stores energy in the isotropic sheet term, with Rayleigh quotient $1/(L_J C_J)$, or $f_{LC} = 17.60\,\mathrm{GHz}$, lowered to 17.49 GHz by the adjacent volume mass. A two-terminal junction cannot carry that pattern. Its 0.001% agreement therefore tests that the two codes assemble the same port operator and nothing more, and the rank-1 shunt of Section 3.3 removes the mode.

The fixture is coarse. Palace's own refinements of it move the qubit from 3.4527 GHz to 3.9528 GHz at $h/2$ (+14.5%) and to 4.1721 GHz at second order on h (+20.8%), and the resonator by +6.1% at $h/2$. Palace's uniform refinement first bisects elements at the metal sheets and then splits every element, so its $h/2$ is a finer mesh than the eight-way split used for the ladders of Section 5; the difference between the two is quantified there. The agreement above is code verification on one discretization; Section 5 supplies the converged numbers.

5 The DeviceLayout Cross-Check

The published designs. The DeviceLayout.jl blog (Peairs and Carson, 2025) reports the initial design ($c = 620\,\mu\mathrm{m}$, $L = 5000\,\mu\mathrm{m}$) at $[4.14, 5.591]\,\mathrm{GHz}$ and the final design ($c = 1217.678\,\mu\mathrm{m}$, $L = 7243.955\,\mu\mathrm{m}$) at $[3.005, 3.982]\,\mathrm{GHz}$, reached by the derivative-free PRIMA optimizer in nine Palace runs "within 1% of our target frequencies". The example source (AWS Center for Quantum Computing, 2025b) sets the targets to $[3.0, 4.0]\,\mathrm{GHz}$, works in the variables $x_1 = \sqrt{620\,\mu\mathrm{m}/c}$ and $x_2 = 5000\,\mu\mathrm{m}/L$, and stops when $\mathrm{err} = \frac{1}{2}\sum_m (f_m - t_m)^2/t_m^2 < 10^{-4}$. Its frequencies are computed with Palace on DeviceLayout's mesh at mesh order 2 and solver order 2, with a first-order absorbing far field and two $50\,\Omega$ readout ports.

Reproduction at DeviceLayout's discretization. Palace, run on DeviceLayout's second-order mesh with the example's configuration file, gives $[4.1683, 5.6132]\,\mathrm{GHz}$ for the initial design and $[3.0164, 3.9956]\,\mathrm{GHz}$ for the final one, 0.68% and 0.40% above the published initial pair and 0.38% and 0.34% above the final pair. The same cells with the PEC-box model used everywhere else differ from the configuration-file runs by at most 0.003%, so the boundary model does not matter here.

The ladder. For each geometry, Palace runs at order 1 on DeviceLayout's first-order mesh h and its uniform refinements $h/2$ and $h/4$ (gmsh's eight-way split, up to 12.6 million unknowns), and at order 2 on h and $h/2$; geode-fem runs at order 1 on h and $h/2$, on the same meshes. Every computed series rises under refinement (Figure 2); this is an observation, not a guarantee. At the edge of a zero-thickness metal sheet the field behaves as $E \sim r^{-1/2}$ (Meixner, 1972), so the eigenfunctions lie

in H^s with $s < 1/2$ near the edge, and the eigenvalue error on uniform meshes scales as h^{2s} at any element order (Boffi, 2010). The asymptotic rate is therefore $O(h)$, and the order-1 ladders show pre-asymptotic orders of 0.61 (qubit) and 0.70–0.73 (resonator).

Extrapolation rule. The *estimate* is the mean of two extrapolations at the asymptotic order 1 in the frequency, $2f_2 - f_1$: one from the order-1 pair ($h/2$, $h/4$) and one from the order-2 pair (h, $h/2$). The *band* spans six values: those two; the same pairs extrapolated at order 1 in the eigenvalue variable $\lambda \propto f^2$, $\sqrt{2f_2^2 - f_1^2}$, which set the low side; and the same pairs at the observed order, which set the high side. The band is an extrapolation range, not a bound, and its high side is kept wide on purpose. For the initial qubit, the circuit-model cap of Section 6 trims the band at 4.477 GHz. The rule was fixed after the static route had been computed: extrapolation at the observed order alone, the rule used before, puts the initial qubit at 4.63 GHz, above the circuit-model cap, and the observed-order values are kept only as the band's high side. The agreement of the two routes is therefore a consistency check, not a blind prediction.

Table 2: Published frequencies, Palace at DeviceLayout's discretization, and the converged estimates with their extrapolation bands, in GHz, from the first refinement family. Percentages are relative to the published value. The estimates and bands are extrapolations, not bounds. Source: `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

	Published	At DeviceLayout's discretization	Converged estimate	Band
Initial qubit	4.14	4.1683 (+0.68%)	4.444 (+7.3%)	$[4.415, 4.677]^*$
Initial resonator	5.591	5.6132 (+0.40%)	5.740 (+2.7%)	$[5.723, 5.817]$
Final qubit	3.005	3.0164 (+0.38%)	3.265 (+8.7%)	$[3.236, 3.471]$
Final resonator	3.982	3.9956 (+0.34%)	4.088 (+2.7%)	$[4.077, 4.139]$

*Under the circuit model of Section 6, the cap of 4.477 GHz trims it to $[4.415, 4.477]$. The second refinement family gives 4.472 $[4.458, 4.510]$.

Result. Table 2 gives the converged estimates. The initial qubit extrapolates to 4.444 GHz and the final qubit to 3.265 GHz, 7.3% and 8.7% above the published values; both resonators extrapolate to 2.7% above theirs. Every published value lies below its band. Measured against the estimates, DeviceLayout's discretization sits 6.2% and 7.6% low on the two qubits and 2.2% and 2.3% low on the resonators, and the first-order mesh h alone sits 22.3% and 26.7% low on the qubits. The estimates rest on Palace's order-1 $h/4$ and order-2 levels, which geode-fem did not run.

Same-mesh agreement on the ladder. geode-fem and Palace agree on the qubit to 10^{-4} % at every shared level of both geometries. On the resonator they differ by 0.03% at h, where Palace adds 4,946 (initial) and 5,944 (final) elements by local bisection of interior boundary tetrahedra, and by 0.0000% at $h/2$, where no bisection occurs and both codes solve the same mesh and, up to the junction model, the same discrete problem. Rerunning one cell with AMS in place of SuperLU, or feeding geode-fem a format round-trip of the mesh, reproduces the same values.

A second refinement family. On the initial geometry we repeated the Palace ladder with a different refinement of the same h mesh: Palace's own uniform refinement, which first bisects the elements at the under-resolved metal sheet. Its $h/2$ qubit is 3.9469 GHz against gmsh's 3.8771 GHz, and on the fixture the two refinements give 3.9528 and 3.8939 GHz. The refinement scheme accounts for 1.5–1.8% of the $h/2$ level, the choice of h mesh for only 0.15–0.43%. The rule applied to the

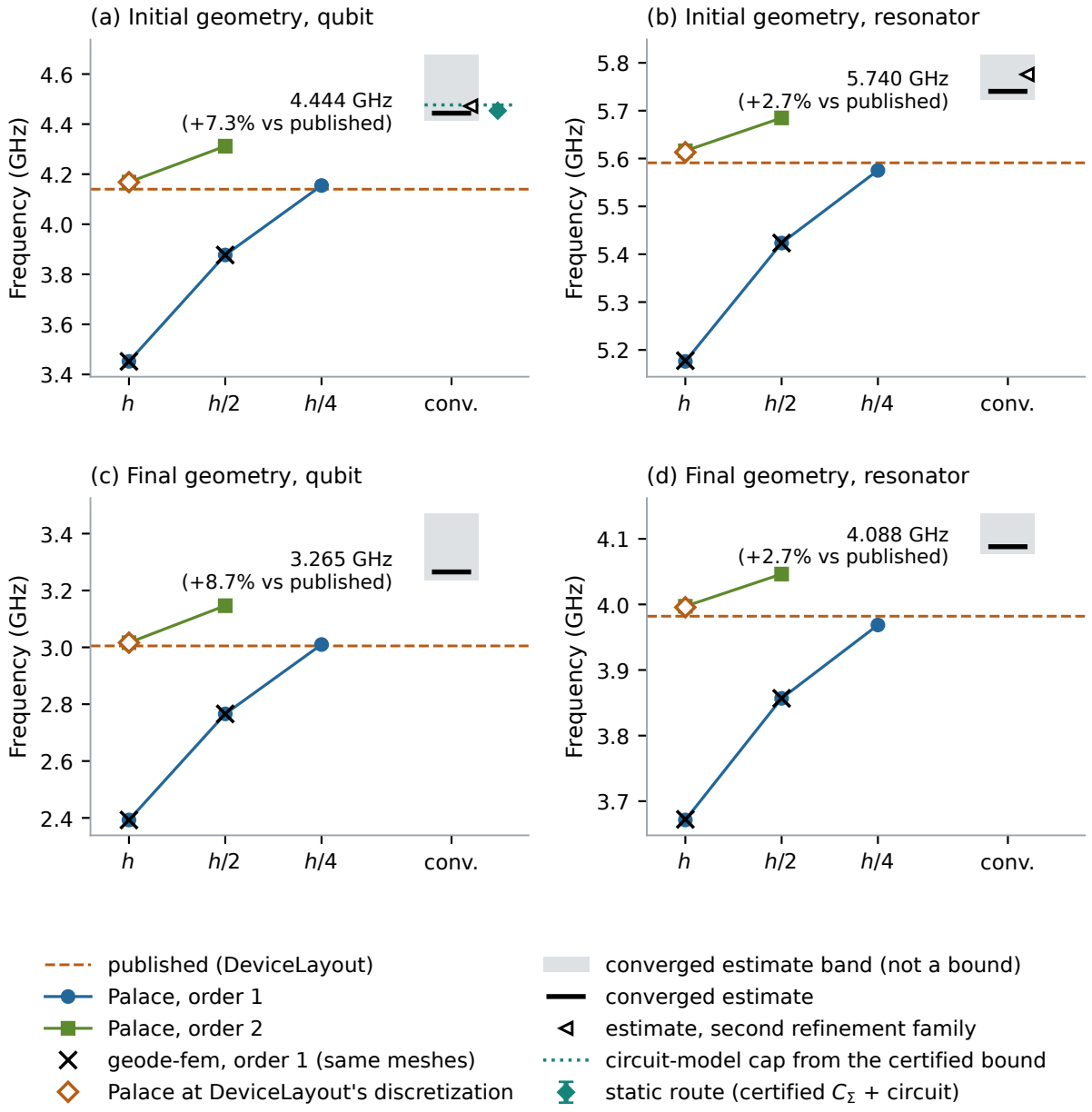


Figure 2: Resolution ladders of the two published DeviceLayout.jl SingleTransmon designs. Palace at order 1 on DeviceLayout's mesh h and its uniform refinements, Palace at order 2, and geode-fem at order 1 on the same meshes (crosses, on top of Palace's order-1 points); the open diamond is Palace on DeviceLayout's own second-order mesh with its configuration file, the discretization of the published values (dashed). The "conv." column is categorical: the converged estimate with its extrapolation band, which is not a bound, and for the initial geometry the estimate of the second refinement family (open triangle). In (a), the static route of Section 6 (certified capacitance bracket read through the junction–resonator circuit) and the circuit-model cap from the certified lower bound. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

second family gives 4.472 [4.458, 4.510] GHz for the qubit, at an observed order of 0.91 against the first family's 0.61, and 5.776 [5.768, 5.800] GHz for the resonator. The two estimates lie 0.64% apart and the second band lies inside the first. We therefore quote the converged initial qubit as 4.444–4.472 GHz, 7.3–8.0% above the published value, and keep the first family's band. Widening the band by the 1.8% level spread would count the extrapolation twice.

Adaptive refinement. An independent check comes from Palace's adaptive mesh refinement, run on the initial geometry from the same h mesh at order 2, with the error indicator summed over the qubit and resonator modes, Dörfler marking at 0.7 and nonconformal refinement (Figure 3). In five solves from 0.95 to 11.0 million unknowns the qubit rises from 4.1683 to 4.4297 GHz. It has not reached a plateau: the last step still moved it by +0.55%, and the run took 3.1 h with a 93 GB peak on 32 ranks. The third solve, at 2.2 million unknowns, already exceeds the finest uniform level (4.3118 GHz at 7.0 million). The finest adaptive level lies 0.32% below the estimate, inside the band and 1.05% below the cap. A three-point fit $f_\infty - CN^{-\alpha}$ through the last three solves gives 4.4552 GHz ($\alpha = 0.80$), 0.26% above the estimate. The fit is exactly determined and ill-conditioned: +0.1% on the finest level moves it by 24 MHz. The adaptive resonator rises more slowly under this marking, and its finest level, 5.6986 GHz, is still below the band; this run does not test the resonator estimate.

A shared-limit diagnostic. Fitting the order-1 ($h/2$, $h/4$) and order-2 (h, $h/2$) series with one common limit and order uses four levels for four unknowns, so it is exactly determined. Table 3 gives it for every ladder in this paper. It lies above the rule's estimate on every first-family series, by 0.5–1.3%, so the rule's point estimates may lean low, and on the initial geometry the second family and the adaptive fit also lie above the first family's estimate. It is a diagnostic, not a correction: on the initial qubit, +0.1% on the order-2 $h/2$ level alone moves it by 20 MHz.

Table 3: The shared-limit diagnostic: the order-1 ($h/2$, $h/4$) and order-2 (h, $h/2$) series of each Palace ladder fitted with one common limit f_∞ and order p, against the extrapolation rule's estimate, in GHz. Exactly determined fits; estimates, not bounds. The step designs are those of Section 7. Source: `benchmarks/transmon_devicelayout_controls/results.toml [shared_limit]`.

	Qubit			Resonator		
Design	p	f_∞	Rule	p	f_∞	Rule
Initial	0.89	4.4802	4.4438	0.81	5.7758	5.7404
Initial, second family	1.02	4.4657	4.4720			
Published final	0.88	3.3022	3.2652	0.85	4.1071	4.0880
Step 2	0.87	3.0389	3.0047	0.84	4.0192	3.9984
Scaling-law step	0.86	3.0418	3.0040	0.85	4.0245	4.0048

Scope. The converged values are estimates for DeviceLayout's geometry under the PEC-box, linear-junction model. On the initial geometry they cover two uniform refinement families and an adaptive run; the final geometry and the re-optimized designs were refined in one family. They are not a statement about a fabricated device.

6 A Static Route: Certified Island Capacitance

The certificate. For the reduced electrostatic problem (island at potential 1, ground and resonator at 0, feedline floating with zero net charge), the potential ϕ satisfies $0 \le \phi \le 1$ by the maximum

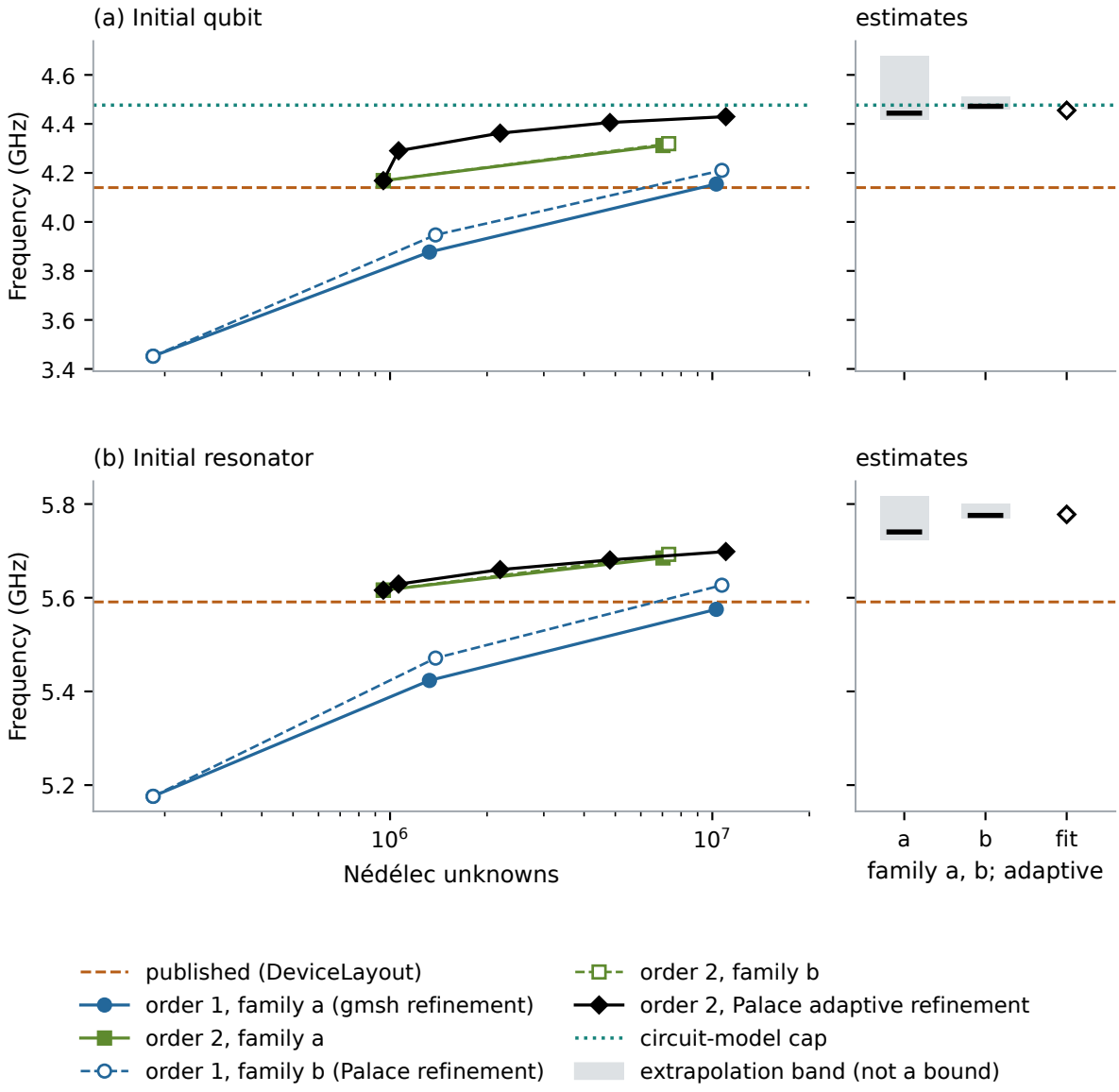


Figure 3: Checks of the converged estimate on the initial geometry. Left: every Palace level against the number of Nédélec unknowns, for the two refinement families of DeviceLayout's mesh h (filled: gmsh's eight-way split, the ladder of Figure 2; open: Palace's refinement) and Palace adaptive refinement from h at order 2 (diamonds). Right: the estimates of the two families with their extrapolation bands, which are not bounds, and the adaptive run's three-point fit, an ill-conditioned estimate. In (a), the circuit-model cap of Section 6. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

principle, and $C_\Sigma = \int \varepsilon \nabla\phi \cdot \nabla\phi$ with $\varepsilon = \varepsilon_0\varepsilon_r$. Every conforming finite-element energy is an upper bound on C_Σ, because C_Σ is the minimum of the energy over admissible potentials. For the lower bound, take any piecewise polynomial field D on the mesh cut along the zero-thickness sheets. Integrating by parts and using $0 \le \phi \le 1$ on every term gives

$$\int D \cdot \nabla\phi \;\ge\; N(D) \;=\; Q_{\text{island}}(D) - |Q_{\text{feed}}(D)| - \|\operatorname{div} D\|_{L^1} - \|[D \cdot n]\|_{L^1(\mathcal{F})}, \tag{3}$$

where Q are the fluxes of D into the two conductors and $[D \cdot n]$ is the normal jump on the interior faces $\mathcal{F}$ off the sheets. Cauchy–Schwarz, $\left(\int D \cdot \nabla\phi\right)^2 \le \int D \cdot \varepsilon^{-1} D \int \varepsilon \nabla\phi \cdot \nabla\phi$, then gives, whenever $N(D) > 0$,

$$C_\Sigma \;\ge\; \frac{N(D)^2}{\int D \cdot \varepsilon^{-1} D}. \tag{4}$$

The bound holds for any D, so solver tolerance can only weaken it. We take D from the P2 potential, equilibrated patch by patch into Raviart–Thomas fluxes of order one to three, with an independent Crouzeix–Raviart flux as a check. The construction is validated against an exact parallel plate, an analytic coaxial bracket and closed-form checks of each term, with mutation tests that reject a grounded-feedline or jump-free certificate. *Certified* in this paper means rigorous for the discrete polygonal geometry of the mesh, with lower bounds floored and upper bounds ceiled. The certificate is evaluated in plain f64, not interval arithmetic; the worst-case summation error is about 10^{-10} relative, far below the rounding. Refinement does not move the polygonal surfaces, so all meshes of one geometry bound the same converged value, and a family bracket is the largest lower bound and the smallest upper bound.

The unmodified device. Uniform refinement of the fixture with P1 and P2 elements gives the upper bounds of Table 4. On adaptively refined meshes the certificate closes the gap: the converged island capacitance of the unmodified initial design X^0 lies in $[77.40, 77.61]$ fF with the scalar $\bar{\varepsilon}$ and in $[79.56, 79.76]$ fF with the sapphire tensor (Figure 4). The certified tensor-minus-scalar difference, computed from the unrounded bounds, is 1.95–2.35 fF, or 2.5–3.1% of the scalar value, and the same relative range holds for the two designs of Appendix A. The committed mesh overstates the tensor C_Σ by at least 56.92 fF, which is at least 71.3% of the converged value, and the scalar one by at least 77.4%.

Table 4: The island capacitance C_Σ (fF) of the unmodified device X^0 on uniform refinements of the committed mesh, and the certified brackets of its converged value from adapted meshes. Each uniform entry is a conforming energy and therefore an upper bound. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml` in `benchmarks/transmon_diffopt/`.

Level	Tetrahedra	P1, scalar	P2, scalar	P1, tensor	P2, tensor
h	133,314	137.706752	90.044141	136.684731	91.717846
$h/2$	1,066,512	101.359397	83.015285	102.375301	85.008425
$h/4$	8,532,096	87.753422	n/a	89.493342	n/a
Certified converged value		$[77.40, 77.61]$		$[79.56, 79.76]$	

From capacitance to the qubit frequency. Reading the certified tensor bracket through the two-node circuit of Section 3.3, over every corner of the C_Σ, L_s and k bands, with the resonator at its converged estimate, gives the initial dressed qubit in $[4.449, 4.458]$ GHz. That range lies inside

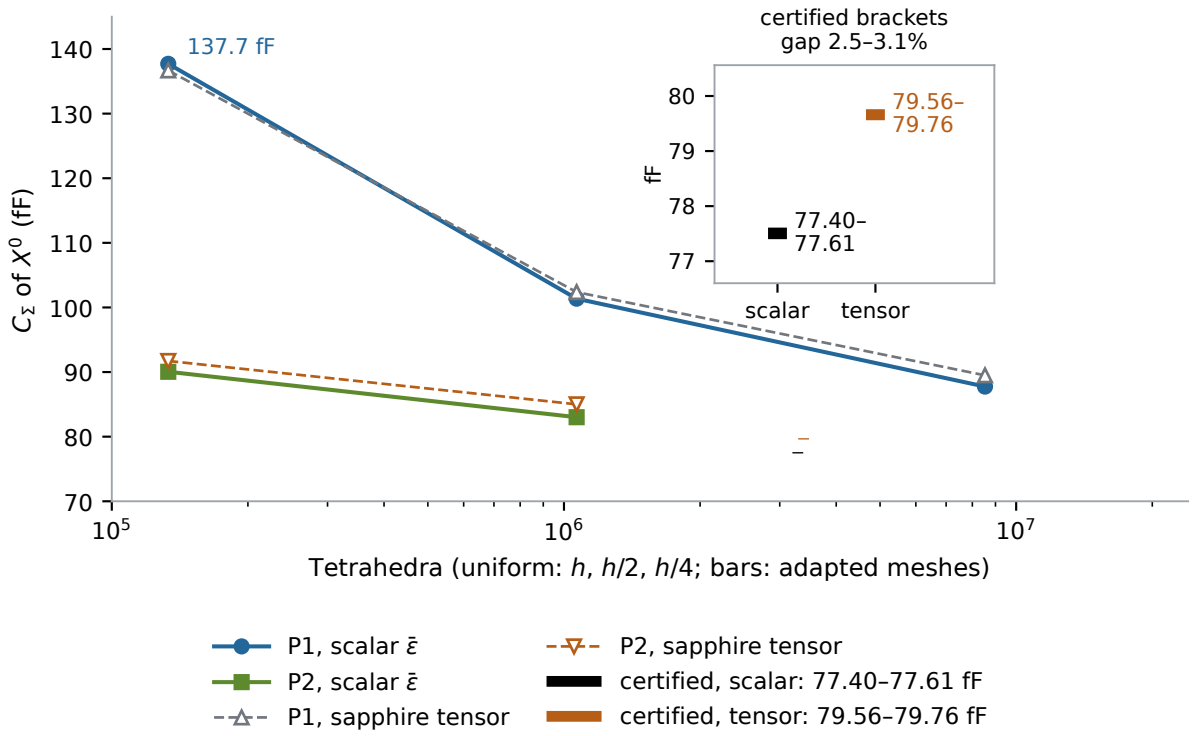


Figure 4: Mesh convergence of the island capacitance C_Σ of the unmodified device under uniform refinement of the committed mesh (Table 4), with the scalar trace-averaged permittivity and with the sapphire tensor. Every marker is a conforming upper bound on the converged value of the committed polygonal geometry. The bars are the certified two-sided brackets from adapted meshes, each drawn at the tetrahedron count of the mesh that sets its lower bound (3,289,543 and 3,388,197), and magnified in the inset on a fF scale; the inset title gives the certified tensor-minus-scalar gap. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml`.

the eigenmode band of Section 5 and inside the range of the two family estimates, and the first family's estimate is 0.22% below its midpoint. Because the extrapolation rule was chosen after this range was known (Section 5), the 0.22% is a consistency check, not an independent prediction. The range is an estimate, because L_s and k are.

A model-dependent cap. Under the circuit model, a series inductance and a coupling to a higher mode can only lower the qubit, so the certified lower bound $C_\Sigma \geq 79.56\,\mathrm{fF}$ caps it at $1/(2\pi\sqrt{L_J(C_\Sigma + C_J)}) = 4.477\,\mathrm{GHz}$. The cap is rigorous only given that model: one island node with the junction $L_J \parallel C_J$, $L_s \geq 0$, coupling only to modes above the qubit, and a quasi-static C_Σ at 4.4 GHz. It also rests on the fixture, which an earlier generator produced from the same DeviceLayout geometry and on which the bracket is certified, not on the meshes of the ladder; the two agree to 0.02% at h. We therefore list it as model-dependent, not certified. The cap trims the high side of the eigenmode band, which comes from extrapolation at the observed pre-asymptotic order (4.677 GHz), and it does not depend on the extrapolation rule. Every computed level lies below it, as do both family estimates and the second family's shared limit (Figure 3, Table 3). The first family's shared limit, 4.4802 GHz, lies 3.6 MHz (0.08%) above it. That excess equals a 0.019% change of the one order-2 $h/2$ level, far inside the 0.15–1.8% mesh and refinement-scheme effects above, and the fit forces one order on two pre-asymptotic series; it is not evidence against the cap.

The adaptive fit, 4.4552 GHz, is as ill-conditioned and is no evidence either way.

7 Gradient-Assisted Re-Optimization and Its Controls

The cross-check says that the published final design does not meet its targets at converged resolution. We repeat DeviceLayout's optimization twice: once against its own discretization, to compare the number of Palace runs with the published run, and once against converged frequencies.

The capacitor_length gradient. DeviceLayout keeps the junction fixed and builds the island upward from it, so a change of c translates every node above the island rigidly, stretches the island band and re-centres the box. geode-fem turns this motion into a node velocity $\partial X/\partial c$, checks on the mesh that every metal outline node in a stretched or blended band lies on an edge the motion maps onto itself (3,146 nodes on the initial mesh), and evaluates Eq. (1) with the sapphire tensor for both modes. The contraction takes 0.44 s. On the initial mesh it matches a central difference of two re-solves of the moved mesh ($\pm 0.5\,\mu$m) to every printed digit: -2.443227×10^{-3} GHz/μm for the qubit and -3.964×10^{-6} GHz/μm for the resonator.

The total_length column. DeviceLayout re-lays the meander when L changes, so no fixed-topology node motion represents it. A transmission-line model holds $f_r(L + \delta)$ constant, with $\delta = 216\,\mu$m from one finite difference of two geode-fem order-1 solves on DeviceLayout meshes ($L = 5000$ and $5250\,\mu$m), and takes $\partial f_q/\partial L = 0$. δ is weakly determined: remesh noise of about 0.15% in either f_r moves it by about $\pm 175\,\mu$m (recomputed from the two committed frequencies), and a second estimate gives about $363\,\mu$m. The model enters only the Jacobian, and every iterate is evaluated on the true objective, so a poorer Jacobian costs steps, not accuracy.

Newton on the high-fidelity objective. DeviceLayout re-meshes every design, so the first-order objective is not smooth in the parameters: from $c = 620$ to $650\,\mu$m the remeshed finite difference of the qubit frequency is -5.23×10^{-3} GHz/μm, against a fixed-topology derivative of -2.28×10^{-3}. The iteration therefore runs Newton directly on the high-fidelity objective F, in DeviceLayout's variables (x_1, x_2), in which the frequencies are close to linear. The c column of its Jacobian is geode-fem's first-order derivative scaled by the measured fidelity ratio $S = F/f_h$ at the same design. Each step costs one geode-fem first-order solve with its gradient (19–25 s) and one high-fidelity evaluation; the one-time δ calibration is setup and is not counted. A first attempt that stepped in (c, L) directly was stopped before its first Palace run finished and restarted in (x_1, x_2).

Table 5: Newton iterates with the geode-fem c column against DeviceLayout's discretization (Palace on its second-order mesh with its configuration file). Step 0 is the initial design. err is DeviceLayout's measure against $[3.0, 4.0]$ GHz; its criterion is 10^{-4}. Source: `benchmarks/transmon_devicelayout_optimize/results.toml`.

Step	c (μm)	L (μm)	f_q (GHz)	f_r (GHz)	err
0	620.000	5000.000	4.1683	5.6132	1.6×10^{-1}
1	1338.338	7137.638	2.8859	4.0465	7.9×10^{-4}
2	1225.346	7223.250	3.0120	4.0009	8.0×10^{-6}
3	1236.360	7224.900	3.0013	4.0011	1.4×10^{-7}

Against DeviceLayout's discretization. Table 5 lists the iterates. The run meets DeviceLayout's criterion at step 2, after 3 Palace runs counting the initial one as DeviceLayout does, against

PRIMA's 9; the three geode-fem solves add about 70 s, against about 4 min per Palace run on 32 ranks. Step 3 reaches err $= 1.4 \times 10^{-7}$ at $(1236.360, 7224.900)\,\mu$m, $+18.7$ and $-19.1\,\mu$m from the published $(1217.678, 7243.955)\,\mu$m. About $+10\,\mu$m of the c offset comes from our reproduction (Palace build, mesher and platform), which gives $[3.0164, 3.9956]$ GHz at the published final design, against the published $[3.005, 3.982]$ GHz. The rest is where PRIMA stopped; its published pair sits 0.17% and 0.45% from the targets, inside its 1% rule. The comparison is with PRIMA's published run count, not with its trajectory, which is not published.

Controls. Two gradient-free controls, run on the same objective and counted the same way, test where the 3 runs come from (Table 6, Figure 5b). The first keeps the L column and replaces the c column by the scaling law $\partial F_q/\partial x_1 = F_q/x_1$, which follows from $f_q \propto 1/\sqrt{C_\Sigma} \propto x_1$ and needs no geode-fem solve. It also meets the criterion in 3 Palace runs, and its first step lands 4.7 times closer (err 1.7×10^{-4} against 7.9×10^{-4}). The second, Broyden's method (Broyden, 1965) started from two forward finite-difference probes, needs 5. In DeviceLayout's variables the problem is close to linear, and the 3-versus-9 compares Newton's method with any of these Jacobians against PRIMA's model-based search; on this problem the shape derivative does not reduce the run count. The two Newton variants share the L column and its δ. Because the Jacobian is diagonal, δ moves only the L component of each step: with $\delta = 0$, 216 and 564 μm (the last a Palace-only value from a Broyden probe) the scaling-law first step has $c = 1196.933\,\mu$m in every case and $L = 7016$, 7141 and $7351\,\mu$m. A fully Palace-only Newton was not run. Run 1 of every method is the same committed cell; repeated on a separately built Palace image, it matches to 6 decimals.

Table 6: Palace runs to DeviceLayout's criterion at its discretization, counting the initial design as run 1. Both Newton variants use the transmission-line L column with the same δ. Broyden's count includes its two probes. PRIMA's first step is not published. Sources: `benchmarks/transmon_devicelayout_optimize/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

Method	c column	Runs	First-step err	Design at criterion (μm)
PRIMA (published)	none	9	n/a	(1217.678, 7243.955)
Newton	geode-fem derivative	3	7.9×10^{-4}	(1225.346, 7223.250)
Newton	scaling law	3	1.7×10^{-4}	(1232.664, 7221.270)
Broyden	two probes	5	6.2×10^{-4}	(1224.847, 7232.447)

Against the converged frequencies. Starting from the published final design, whose converged estimate is 3.265 and 4.088 GHz, each Newton step uses the converged estimate and the geode-fem gradient at the current design, and each new design is evaluated by a full ladder (Palace at order 1 on h, $h/2$, $h/4$ and order 2 on h, $h/2$, geode-fem at order 1 on h, $h/2$) with the rule of Section 5. Table 7 gives the two steps. The first overshoots the qubit by -1.09%; the second lands at $+0.16$% and -0.04% of the targets, with both targets inside the bands. On both ladders geode-fem and Palace agree on the qubit to 2×10^{-4} %. One scaling-law step from the same start, evaluated by the Palace half of the ladder, lands 3.6 and 8.2 μm short of step 2 and at $+0.13$% and $+0.12$% of the targets: one gradient-free step does what the two geode-gradient steps did.

The converged design. At converged resolution the 3.0/4.0 GHz design needs a substantially longer capacitor than the published one: the point estimate is $c \approx 1446\,\mu$m, $+19$% ($+228\,\mu$m), with a resonator 2.4% longer ($+172\,\mu$m). Mapped linearly through $dF/dc \approx S\,\partial f_h/\partial c = -9.05 \times$

Table 7: Newton steps against the converged frequencies. Each design is evaluated by a full resolution ladder (the scaling-law step by its Palace half); the converged values are estimates with extrapolation bands, not bounds. Sources: `benchmarks/transmon_devicelayout_optimize/results.toml`, `benchmarks/transmon_devicelayout_controls/results.toml` and, for the published final design, `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

Design	c (μm)	L (μm)	f_q [band] (GHz)	f_r [band] (GHz)
Published final	1217.678	7243.955	3.265 [3.236, 3.471]	4.088 [4.077, 4.139]
Step 1	1481.657	7407.945	2.967 [2.938, 3.188]	4.004 [3.993, 4.055]
Step 2	1446.078	7416.352	3.005 [2.979, 3.221]	3.998 [3.987, 4.067]
Scaling-law step	1442.479	7408.181	3.004 [2.977, 3.235]	4.005 [3.994, 4.055]

10^{-4} GHz/μm, the slope at the step-1 design where step 2 started, and centred on the step-2 estimate of 3.005 GHz, the step-2 qubit band corresponds to c between about 1418 and 1685 μm, +16% to +38%. The point estimate may lean low: the shared limit of Table 3 puts step 2 at 3.039 and 4.019 GHz, +1.3% and +0.5% over the targets, which would call for a longer capacitor still; it is an ill-conditioned diagnostic and inside the band. Step 2 therefore lands on the targets within the band; it is not a verified hit of 3.0/4.0 GHz at convergence. Each ladder took 1.4 and 1.6 h of Palace wall time, dominated by the order-2 $h/2$ cell.

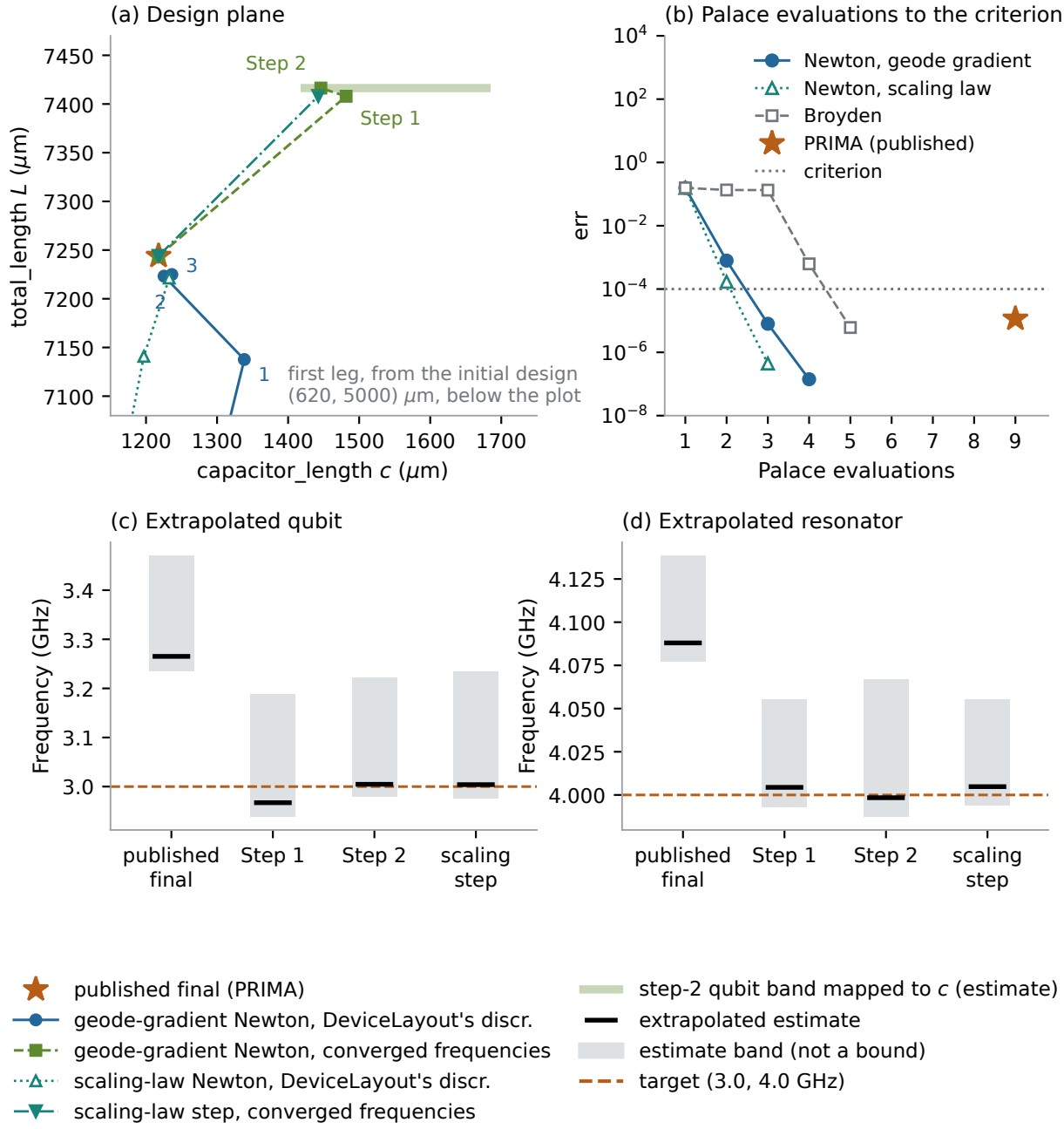


Figure 5: The re-optimization and its controls. (a) The design plane near the optimum: the published final design, the Newton iterates against DeviceLayout's discretization with the geode-fem and the scaling-law c column (numbered: geode-fem; the first steps start from the initial design, off the plot), and the steps against the converged frequencies, with the step-2 qubit band mapped linearly to capacitor_length (an estimate). (b) DeviceLayout's err against the number of Palace evaluations at its discretization for the two Newton variants and Broyden's method; PRIMA's published end point is plotted at its 9 runs, with err computed from the published frequencies, since its trajectory is not published. This counts Palace evaluations, not total cost: each geode-gradient Newton point also costs one geode-fem solve with its gradient, and both Newton variants share the one-time δ calibration. (c, d) Extrapolated estimates and bands of the published final design and the converged-target steps, against the targets. Sources: the `optimize`, `controls` and `crosscheck` artifacts under `benchmarks/transmon_devicelayout_*/` (Table 9).

8 The Sensitivity Layer

For a solve $A(\varepsilon, X)\, x = b(X)$ and an observable $g(x)$, the discrete adjoint $A^{\mathsf{T}}\lambda = \partial g/\partial x$ gives

$$\frac{dg}{d\varepsilon_k} = -\lambda^{\mathsf{T}} \frac{\partial A}{\partial \varepsilon_k} x, \qquad \frac{dg}{dX} = \lambda^{\mathsf{T}} \frac{\partial b}{\partial X} - \lambda^{\mathsf{T}} \frac{\partial A}{\partial X} x, \tag{5}$$

and in the complex driven case $2\,\mathrm{Re}[\cdot]$ of the same contractions with the transpose solve, since the pencil is complex-symmetric. The element Jacobians are exact, computed by hand-written forward-mode dual-number twins of the production kernels, and each gradient costs one forward and one adjoint solve on one factorization. Table 8 lists the four modules. Each has a test against a central difference of the full pipeline (perturb, re-assemble, re-solve, re-extract) and mutation tests: the conjugation error A^{H} for A^{T} and a dropped term both fail. In the driven shape case the source $b = \mathrm{i}\omega\mu_0 \int N \cdot J$ depends on the geometry, and omitting $\partial b/\partial X$ fails the check.

Table 8: The discrete-adjoint sensitivity layer. Each cell: one forward and one adjoint solve on one factorization, exact element Jacobians, and the observed agreement with a central finite difference of the full pipeline. Modules in `crates/geode-core/src/`.

Operator	Parameter	Module	FD rel-err
scalar electrostatic (SPD)	material $\partial/\partial\varepsilon$	`adjoint`	$\sim 3 \times 10^{-8}$
scalar electrostatic (P1)	geometry $\partial/\partial X$	`shape`	$\sim 1 \times 10^{-9}$
driven $H(\mathrm{curl})$	material $\partial/\partial\varepsilon$	`driven::adjoint`	2.3×10^{-5} (worst region)
driven $H(\mathrm{curl})$	geometry $\partial/\partial X$	`driven::shape`	2.2×10^{-9}, 1.2×10^{-8} (two maps)

For the capacitance $C = \phi^{\mathsf{T}} K(X) \phi$ the stored energy is stationary in the solved potential, so the implicit term vanishes and $dC/dX = \phi^{\mathsf{T}}(\partial K/\partial X)\phi$, the classical self-adjoint result (Park, 2019). The implementation keeps the general adjoint form, and its test measures the implicit part below 10^{-8} of the total. Composed with $E_C = e^2/(2(C_\Sigma + C_J))$, this gives $\partial E_C/\partial X$. On a parallel-plate fixture Newton reaches a target E_C in one step, as expected for its affine parameterization, and a fresh solve reproduces the target to 1.4×10^{-15}. Appendix A applies the same identity, in a native-f64 P2 implementation outside the Burn tape, to a three-parameter electrostatic optimization of the real device. The eigenfrequency derivative of Section 7 is the analogous stationarity result for the eigenproblem.

9 How geode-fem Was Designed and Built

geode-fem implements the whiteroom L1–L4 operator specification of Crutcher Dunnavant (Dunnavant, 2026). He also gave design input on the solver's operator layer and is the author of 36 commits, made in June 2026, that migrated the code to Rust 2024, added the Metal (Apple GPU) backend, restructured the core crate and the examples into their present layout, and created the geode-util crate. The code, the experiments and the drafts of this paper were otherwise produced by frontier AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). Codex wrote an earlier full draft of this paper and an earlier redesign of its figures, and wrote the P2 shape-derivative code and evidence behind the electrostatic design loop of Appendix A. The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by a shared Squad room.

Nearly every change was reviewed by a separate agent session: 48 of the 50 pull requests had a review or comment from a separate session, and once verifier scripts were added, the reviews of each paper revision recomputed the numbers from committed artifacts, reproduced runs and

mutation-tested the verifier pins. Several claims in this paper were corrected or dropped because of those reviews. Three examples follow. The 4.14 GHz of the SingleTransmon example was first read as a design target and set a capacitance anchor; checked against the primary source, it is the frequency of the unmodified initial geometry, and the designs aimed at it are now supporting material (Appendix A). A capacitance comparison that mixed the scalar-$\bar{\varepsilon}$ and sapphire-tensor models was replaced by certified brackets for each model and a certified gap between them (Section 6). The agreement of the static route with the eigenmode estimate was first presented as an independent check; the extrapolation rule had been fixed after the static route was known, so it is reported as a consistency check (Section 5).

Counts from repository history, before this revision: the paper and its benchmark artifacts changed through 50 merged pull requests from 2026-07-14 and 94 commits. Of these commits, 91 carry a Claude co-author trailer (Opus 75, Fable 10, Sonnet 6); the other 3 were made through Codex, which adds no trailer, and contain about 10% of the added text lines. The human author directed the research, made every scoping and claim decision, and takes responsibility for the content.

10 Reproducibility

Every result in this paper is read from a committed, scripted artifact.[1] Table 9 lists them. The versions are geode-fem commit `3174015` for the same-mesh agreement, with the commit of each later result recorded in its artifact, Palace `fba6a5b`, DeviceLayout.jl v1.15.0 and Burn 0.21. The DeviceLayout ladders and the adaptive run ran on AWS Graviton4 hosts (`r8g.8xlarge`, 32 vCPU, and `r8g.4xlarge`, 16 vCPU); the capacitance brackets ran on a local 28-core, 96 GiB workstation. The Palace runs consume an MSH 2.2 conversion of each mesh, because MFEM rejects DeviceLayout's multi-block MSH 4.1 file; the conversion keeps every node and physical group and reproduces the eigenvalues bit for bit. The figure scripts in `figures/src/` read the artifacts directly.

Table 9: Artifacts behind each result, relative to the repository root. The generated artifacts are pinned by regression tests.

Result	Artifact
Same-mesh agreement (Table 1)	`benchmarks/transmon_eigen/results.toml`; Palace cells under `reference/fixtures/transmon_palace/`
Cross-check (Table 2, Figure 2)	`benchmarks/transmon_devicelayout_crosscheck/`; Palace cells under `reference/fixtures/transmon_palace/results_crosscheck_1092/`; meshes from `reference/julia/generate_transmon_param.jl`
Second refinement family, adaptive run, shared limits, controls (Figure 3, Tables 3, 6)	`benchmarks/transmon_devicelayout_controls/`; Palace cells under `reference/fixtures/transmon_palace/results_controls_1123/`
Capacitance convergence and brackets (Table 4, Figure 4)	`benchmarks/transmon_diffopt/c_sigma_convergence.toml`, `c_sigma_bracket.toml`, `c_sigma_bracket_tensor.toml`, `c_sigma_bracket_anchor_93p95.toml`
Circuit terms L_s and k	`benchmarks/transmon_loop_inductance/results.toml`, `benchmarks/transmon_coupling/results.toml`
Re-optimization (Tables 5, 7, Figure 5)	`benchmarks/transmon_devicelayout_optimize/`; Palace cells under `reference/fixtures/transmon_palace/results_optimize_1092/`
Electrostatic design loop (Appendix A)	`benchmarks/transmon_diffopt/refined_optimization/`
Performance (Appendix B)	`benchmarks/transmon_bench_cpu/`, `benchmarks/gpu_driven_scaling/`

[1] The geode-fem source is public at https://github.com/rjwalters/geode-fem. Archival DOI to be added at deposit.

11 Discussion

What is established, and how. Table 10 sorts the results by the kind of evidence behind them. The brackets are the only bounds. The cap follows from a bracket only under a circuit model. The converged frequencies and the re-optimized design are extrapolations with stated rules and bands, and the static route rests on circuit terms that are themselves extrapolated. What stands without any extrapolation rule is the finest computed levels, already 4.1% and 4.7% above the published qubit values, and on the initial geometry the adaptive run's 4.4297 GHz, 7.0% above.

Table 10: Evidence tiers. Certified: a guaranteed bound for the meshed polygonal geometry. Model-dependent: a rigorous consequence of a certified bound under a stated circuit model. Measured: a committed solve on a stated discretization. Estimated: an extrapolation with a stated rule, not a bound. Consistency: an agreement that is not independent evidence.

Tier	Results
Certified	C_Σ of X^0: $[77.40, 77.61]$ fF (scalar), $[79.56, 79.76]$ fF (tensor); tensor-minus-scalar gap 2.5–3.1%
Model-dependent	The initial-qubit cap of 4.477 GHz: two-node circuit, quasi-static C_Σ, bracket certified on the fixture's polygonal geometry, not on the ladder meshes
Measured	Same-mesh agreement (Table 1); Palace at DeviceLayout's discretization within 0.7% of the published pairs; every ladder and adaptive level, the finest uniform ones already 4.1% and 4.7% above the published qubit values; same-mesh agreement on the ladders; the gradient check; 3 Palace runs to DeviceLayout's criterion with either Newton c column, 5 with Broyden's method
Estimated	Converged frequencies and bands (Table 2), the initial qubit 4.444–4.472 GHz over two refinement families; the adaptive fit and the shared limits; the static-route qubit range; L_s and k; the converged design $c \approx 1446\,\mu$m and its +16 to +38% range
Consistency	The two routes agree to 0.22% on the initial qubit; the extrapolation rule was fixed after the static route was known, so this is not a blind prediction
Not claimed	Agreement with measurement; a converged hit of 3.0/4.0 GHz; a global optimum; a reduction of the Palace run count by the shape derivative; eigenmode-participation (EPR) derivatives; a speed advantage over Palace

What the derivative contributes. On this two-parameter problem, in variables chosen to make the frequencies nearly linear, a scaling law is as good a Jacobian as the shape derivative, and the shape derivative does not reduce the number of Palace runs. What this paper shows for the derivative is that it is exact against finite differences on the mesh and cheap (0.44 s for the contraction). Problems with more parameters, or without a scaling law per parameter, are where a field-solve derivative would have to show its value; we have not tested one.

Threats and limitations. Agreement between two codes is evidence against implementation error, not against shared modeling choices, and the 17.49 GHz port-sheet mode is an example of a shared artifact; the rank-1 shunt removes it in geode-fem, and Palace still has it. The extrapolations assume a rate the sheet-edge singularity sets, their high sides are wide, and only the initial geometry was refined in a second family and adaptively; the adaptive run had not reached a plateau. The circuit route uses a bracket certified on a different mesh of the same initial geometry, and L_s and k were extrapolated from two or three discretizations. The re-optimization's transmission-line model rests on one weakly determined δ, which moves only the L component of each step and does not affect the iterates' evaluation. The PRIMA comparison counts runs only. The electrostatic certificate is evaluated in f64 and its code is validated by tests, not formally verified. The model is lossless and linear: the junction is $L_J \parallel C_J$, the readout ports are open in the ladders, and the

metal is a set of zero-thickness sheets. Only one device is studied. The eigenmode branch is not yet differentiated with respect to participation ratios or Kerr couplings; an interior eigensolve that is robust at the physical shift is the missing piece, and a projected Jacobi–Davidson iteration (Liang et al., 2026) with the eigenpair derivatives of Nelson (1976) is the planned route.

12 Conclusion

The published frequencies of the DeviceLayout.jl SingleTransmon example are reproduced by Palace at the example's own discretization, and the extrapolated qubit-frequency estimates are 7–9% above them; geode-fem reproduces Palace on every level both codes ran, and an adaptive Palace run and a second refinement family fall inside the extrapolation band. The zero-thickness sheet edge limits uniform refinement to $O(h)$ at any element order, and DeviceLayout's default discretization sits 6–8% low on the qubits. A certified bracket on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model. Newton steps in DeviceLayout's variables reach its criterion in a third of the published Palace runs, with geode-fem's eigenfrequency gradient or a scaling law in the Jacobian alike, and at converged resolution they move the optimized design to a capacitor about 19% longer, within a band of 16–38%. For design examples that are meant to be copied, the discretization is part of the result and belongs in the report beside it.

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A Supporting Material: A Certified Electrostatic Design Loop

This appendix reports an electrostatic optimization of the real device whose interest is its acceptance test, not its target. The target, an island capacitance of 93.954 fF, was obtained by reading the published initial-geometry frequency of 4.14 GHz as a design target through the linear junction circuit. Since 4.14 GHz is the frequency of the unmodified initial design, the target has no design meaning; the runs remain a measured example of how much a target reached on an optimization mesh is worth after refinement.

Method. Ten cycles of adaptive refinement turn the committed mesh into a 448,279-tetrahedron optimization mesh. Three junction-pinned harmonic morph fields change the island length, the pad width and the cutout gap. The gradient is the stationary-energy shape derivative of Section 8 for the floating-feedline C_Σ, evaluated with P2 fields and exact quadrature on each affine tetrahedron, in native f64 outside the Burn tape. The problem is underdetermined (three parameters, one target), and the accepted steps are those that reduce the freshly solved residual and keep the minimum element-volume ratio above 0.25. The design is scalar-$\bar{\varepsilon}$.

Result. The optimizer raises C_Σ from 78.387918 to 93.953668 fF in four accepted steps; at the final design all three sensitivities agree with central differences of fully re-solved geometries to 4.16×10^{-7} (rounded up). Seven held-out refinement rounds, none of which feed the optimizer, lower the conforming value to 92.998560 fF at 3,388,197 tetrahedra, and the certificate of Section 6 on the same meshes brackets the converged capacitance of the design in $[92.75, 93.00]$ fF with the scalar model and $[95.34, 95.58]$ fF with the sapphire tensor. The target reached on the optimization mesh is lost by about 1% under refinement, and the certificate bounds the loss. A second run aimed past that bias (target 95.112986 fF) gives $[93.89, 94.14]$ fF scalar and $[96.51, 96.75]$ fF tensor. Palace with second-order elements on the exported adapted mesh agrees with geode-fem's C_Σ to 2.5×10^{-6}, which tests the two assemblers on one discretization and says nothing about the discretization error.

What it shows. A gradient loop converges to the target it is given on the mesh it is given, and a finite-difference check or a fresh solve on that mesh says nothing about the discretization error. The held-out conforming series and the equilibrated-flux certificate together give a two-sided acceptance test about 0.25 fF wide, independent of the optimizer. The same certificate, with the tensor, supplies the bracket of Section 6.

B Performance Context

The timings below qualify the cost of the solvers; none measures the re-optimization or ranks the two codes. The CUDA backend (burn-cuda 0.21) is f32-only today, because the underlying cubecl runtime currently disables f64,[2] so every GPU result is mixed-precision-qualified.

CPU eigenmodes. On one otherwise idle EC2 `m6i.4xlarge`, with every request verified to return the six reference modes of Table 1, Table 11 gives median wall seconds and measured core-seconds. Two of the geode-fem requests are tuned using the reference spectrum, and Palace solves about 17% more unknowns from the same input mesh, so these are workload comparisons, not kernel speedups. With eight threads or ranks, Palace without field output is fastest; with one, the geode-fem routes take less time, but geode-fem writes no fields and has no counterpart to Palace's error estimator.

[2] An upstream constraint of the burn/cubecl stack at version 0.21 (Tracel AI, 2024): the CUDA runtime reports no f64 support, so the f64 code paths are not exercised on the GPU.

Table 11: Mode-verified CPU timings on one m6i.4xlarge: median wall seconds / measured core-seconds, with one or eight threads (geode-fem) or MPI ranks (Palace). All rows contain the six reference modes of Table 1. Three repeats per cell except Palace one-rank Save=0 (one run). The two requests tuned on the reference spectrum are marked †. Palace always requests six modes above 4.5 GHz. Source: `benchmarks/transmon_bench_cpu/results_like_for_like_m6i.toml`.

Request	Extra modes	1 thread/rank	8 threads/ranks
geode 30 at 4.5 GHz†	24	34.45 / 34.44	29.08 / 40.71
geode 6 at 20 GHz†	0	33.21 / 33.20	27.26 / 38.87
geode port-aware 8 at 4.5 GHz	2	67.26 / 67.25	51.12 / 91.72
Palace, Save=6	0	186.84 / 186.82	33.35 / 266.66
Palace, Save=0	0	113.00 / 112.98	23.64 / 188.89

Driven solves and memory. On assembled first-order driven problems, geode-fem's AMS preconditioner (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009) reduces one-core solve time relative to Jacobi by 3.4–4.2× across 25,695–897,650 edges, but Palace's AMS is faster at scale (53.3 against 181.3 s at the largest size). On a lumped-port cube family up to 25,695 edges, geode-fem's GPU-f32 matrix-free solve is slower than both assembled CPU configurations at every measured size and reaches parity with CPU matrix-free only at the largest. A geode-fem direct eigensolve at 1,157,564 unknowns completed in 565.5 s at 92.2 GB peak memory, on a different host from Palace's 423.12 s run; the direct factorization's memory, not a controlled crossover, is what that shows, and geode-fem's direct solves at the ladders' $h/2$ peaked at 94 and 136 GB, which is why Palace ran the largest cells (up to 12.6 million unknowns).

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